Where Should We Land?

Hardware Beacon Network vs. Software Visibility Simulation
NASA Space Apps Challenge 2026: Lunar Landing-Site Challenge

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October 2026

1 The Problem

The lunar south pole is a moving target. The Sun and Earth rise only about 2–7° above the horizon, so crater rims easily block both. Earth follows a two-week cycle: visible for two weeks, hidden for two weeks. Long shadows sweep around the pole, so a site that is sunny today can be dark next week.

A landing site is a point in time, not just a point on a map.

NASA's reference requirements for Artemis III, from its briefing for the International Space Apps Challenge [1], are:

- Land within 6° of the south pole.
- Surface slope under 10° over 100 m.
- Direct-with-Earth communication. **Assume no relay is available.**
- Lighting for power and for the approach.

2 Approach A: Hardware Beacon Network

Beacons sit on the surface around the work area, and the vehicle talks to them as a mesh network. A hub beacon on a sunlit spot links to a satellite in lunar orbit, and the satellite relays everything to Earth (Figure 1). For the hackathon, the deliverable would be a tabletop mesh demo, for example with ESP32 radios.

3 Approach B: Software Visibility Simulation

This approach takes NASA's terrain data and Sun and Earth positions, and computes for each candidate site when it has sunlight and when it can see Earth, hour by hour. It then scores and ranks the sites and shows them on a web map (Figure 2).

User inputs: mission start date and duration (14–60 days), minimum Earth-visibility percentage, and the longest dark period the battery can survive.

Output: a ranked list of landing sites with the reason each one ranks where it does, plus a per-site timeline of power, communications, and both together.

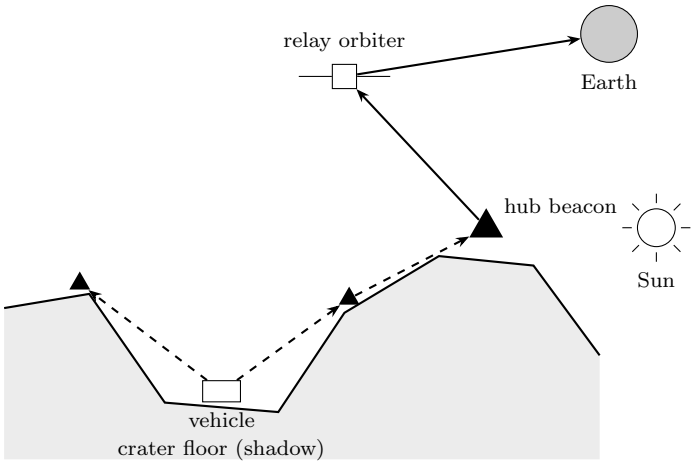


Figure 1: Approach A. The vehicle reaches Earth through surface beacons, a hub beacon, and a relay orbiter. Dashed lines are surface mesh links.

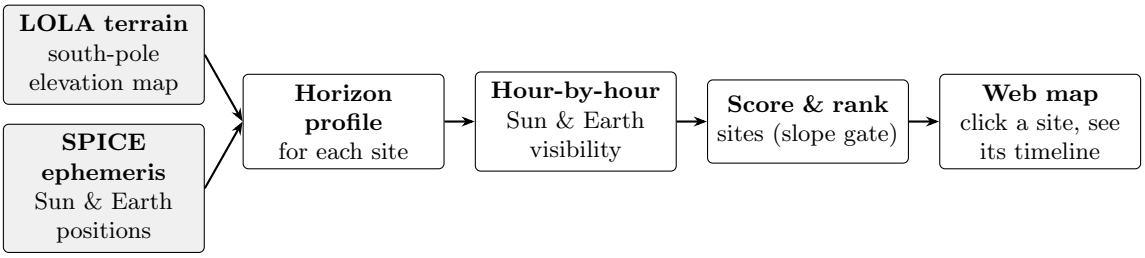


Figure 2: Approach B pipeline. Shaded boxes are NASA data inputs.

4 Comparison

Table 1 compares the two approaches. ✓ means strong, ✗ means weak, and ∼ means partial.

5 The Core Flaw in Approach A

Every beacon is a small lander. It needs sunlight, so it needs a sunlit spot. The hub needs a clear view of the relay or Earth. And someone has to deliver and place the beacons *before* the vehicle arrives.

So Approach A does not replace Approach B. It multiplies the site-selection problem and still needs B to solve it (Figure 3).

6 Where Approach A Does Belong

The idea is not bad. It is aimed at a different problem.

- **Relays are real.** Far-side CLPS missions such as ispace-U.S. APEX 1.0 (Schrödinger Basin) use relay satellites for surface operations [4]. That is a full extra spacecraft, though, not a way to choose a landing site.
- **Surface meshes make sense after landing**, for example to keep a rover connected inside a shadowed crater.

Criterion	A: Beacon network	B: Simulation
Answers "where should we land?"	✗ Only works after landing	✓ Outputs ranked sites
Fits NASA's "no relay" assumption	✗ Depends on a relay orbiter	✓ Models direct-to-Earth
Uses NASA open data	✗ Barely	✓ LOLA terrain, SPICE
Buildable in 48 h	∼ Tabletop toy only	✓ MVP is realistic
Proof judges can check	✗ A bench demo is not the Moon	✓ Match published illumination maps
Cost	∼ Radios, boards, power	✓ Laptops only
Fits our embedded skills	✓ Our home turf	∼ Python + geometry
Novelty	✓ Unusual	∼ Common idea, must stand out
Strong on	**2 of 8**	**6 of 8**

Table 1: Head-to-head comparison.

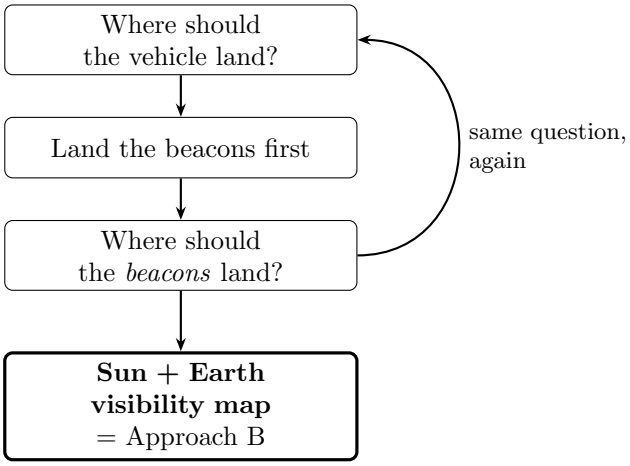


Figure 3: Approach A loops back to the same question, and the only way out is Approach B.

- **Its natural home is a lunar-communications challenge.** NASA's 2027 Human Lander Challenge is about exactly that, but it is open only to U.S.-based teams [3].

We keep it as one "future work" point in our final pitch: *once our tool picks the site, a surface relay network can extend coverage into shadowed craters.*

7 What Approach B Computes

1. Horizon profile, once per site. From the site height h_0, for each azimuth θ (every 1°), march outward over terrain samples at distance d_i. The $d_i^2/2R$ term is the drop due to the Moon's curvature.

$$H(\theta) = \max_i \arctan \frac{h_i - h_0 - d_i^2/2R}{d_i}, \qquad R = 1737.4 \text{ km}$$

2. Visibility, every hour t. Sun and Earth azimuth and elevation come from SPICE.

$$\text{Sun visible} \iff \text{el}_\odot(t) > H\big(\text{az}_\odot(t)\big), \qquad \text{Earth visible} \iff \text{el}_\oplus(t) > H\big(\text{az}_\oplus(t)\big)$$

The horizon profile is computed once, so each hour is just a lookup.

3. Score. Only sites that pass the slope and latitude limits are scored. f is the fraction of hours in the mission window T, and g is the longest dark gap.

$$S = w_1 f_{\text{sun}} + w_2 f_{\text{earth}} + w_3 f_{\text{both}} - w_4\, g/T$$

8 Scope for 48 Hours

Build (MVP):

- South-pole terrain tile.
- Horizon profiles on a grid of candidate sites.
- Sun and Earth positions via SpiceyPy.
- Visibility timelines and scores.
- 2D map of the scores; clicking a site shows its timeline.

Stretch: user-set weights (power vs. communications), a 3D terrain view of the top sites, and an exportable site report.

Cut: the rocket descent animation, the launch-site input (it does not affect the landing site), and the beacon hardware.

How we stand out: judges will see other landing-site maps. We win on the *time* dimension, meaning when each site works and for how long. We also show that our sunniest sites match published illumination maps, for example the rim of Shackleton crater.

9 Recommendation and Timeline

We recommend Approach B, for three reasons:

1. It answers the question asked: where to land.
2. It matches NASA's assumption of direct-to-Earth communication with no relay.
3. It can be built and checked in 48 hours using NASA's own data.

Timeline [2]:

- **Now to October 28:** lock the approach, download the terrain data and SPICE kernels, and get the horizon profile working for one site.
- **October 28:** full challenge statements and datasets are released. Re-check our scope against them.
- **November 14–15:** the hackathon. Build the MVP first, then the stretch goals.

Decision needed: agree on Approach B, with Approach A kept as one future-work point.

References

[1] NASA, *Artemis Overview, 2025 International Space Apps Challenge*, NTRS 20250009323. https://ntrs.nasa.gov/api/citations/20250009323/downloads/ArtemisOverview2025IntSPAPPSChallenge.pdf

[2] NASA Space Apps Challenge 2026. https://www.spaceappschallenge.org/2026/

[3] NASA, *NASA Opens 2027 Human Lander Challenge for Lunar Communications*. https://www.nasa.gov/?p=1051652

[4] ispace-U.S., *Draper Awards ispace an Additional $7.7 Million as part of the NASA CLPS CP-12 Task Order*. https://ispace-us.com/2025/03/draper-awards-ispace-an-additional-7-7-million-as-part-of-the-nasa-clps-cp-12-task-order